

HCMC University of Technology and Education
Faculty of Mechanical Engineering
Department of Mechatronics



**AI PROJECT: BIRD SPECIES
CLASSIFICATION BY SONG**

Lecturer: Dr. Tran Vu Hoang

Members

Order	Name	ID
1	Dinh Duc Duy	22134001
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ThuDuc, 26th December

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I. Introduction

1.1. The reason for this project

The classification of bird species through their songs is an area of active research with rapid advancements and numerous applications in bioacoustics, ecology, environmental studies, and the conservation of endangered bird species. The world is facing a biodiversity crisis due to human activities such as environmental pollution, climate change, and habitat destruction *Figure 1.1*. Monitoring and conserving animal species, which serve as indicators of biodiversity, is therefore crucial.

Birds, in particular, are vital indicators of environmental changes, including habitat shifts, migration patterns, pollution levels, and outbreaks of diseases. Conserving bird populations is essential for understanding ecological changes and protecting global biodiversity.

East Africa is home to many bird species that play critical roles in ecosystems and contribute to the region's cultural and economic values. However, monitoring and conserving these species are challenging due to vast and diverse landscapes, rugged terrains, and the complex acoustic signatures of their songs. Traditional monitoring methods are often costly, time-consuming, and limited in data coverage, particularly for rare and threatened species. These limitations can be addressed through **Passive Acoustic Monitoring (PAM)** in *Figure 1.2*. By combining PAM with advanced analytical tools, conservationists can sample at larger spatial scales, achieve higher temporal resolution, and better understand the relationships between restoration efforts and biodiversity.

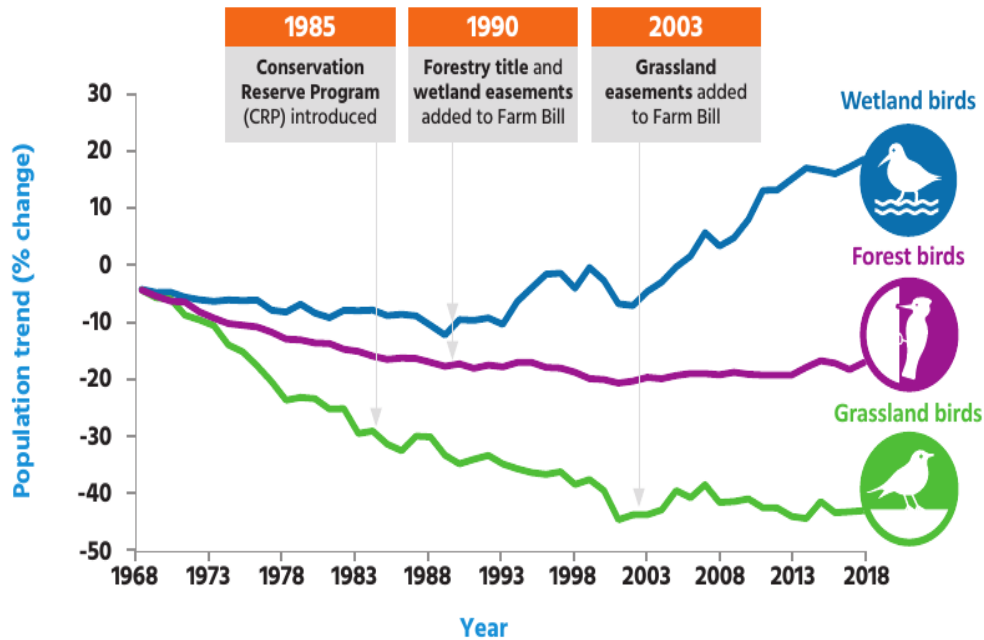


Figure 1.1: Average population trends for suites of bird species dependent on major habitat types[1]



Figure 1.2: Passive Acoustic Monitoring device

1.2. Project aim

The goal of this project is using Deep Learning to develop robust models capable of accurately identifying and classifying East African bird species based on their songs.

1.3. Limitation

Due to time constraints, this project will exclude certain advanced techniques such as transfer learning, generative adversarial networks (GANs), and additional dataset collection. The focus will remain on optimizing available resources to achieve reliable bird species classification.

1.4. Working table

Time	Duy	Trí
6/11 – 20/11	Find most used model currently	Topics and related thesis
20/11 – 5/12	Data preprocessing	Audio conversion
5/12 – 10/12	Deploy model	Model evaluation
10/12 – 26/12	Find best model + Word	Word + PowerPoint

II. Related work

2.1. Challenges in birdsong classification

Bird species classification based on songs poses numerous challenges in real-world scenarios. One prominent issue is noise, which can originate from the environment or the recording device itself, degrading the quality of the audio data. Additionally, the high variability in bird songs, caused by differences in pitch, tempo, and patterns among species, complicates the classification process. Other significant challenges include the short duration of certain calls and overlapping calls from multiple species, making it difficult to distinguish between them. Another critical issue is data imbalance, where certain bird species are overrepresented while others are underrepresented in datasets. This imbalance can bias the classification model towards frequently occurring species, reducing its ability to accurately classify rare or endangered bird species.

2.2. General method

The general method for bird species recognition is illustrated in *Figure 2.1*. Initially, the songs of various bird species are captured using an audio recording system. The captured signals are then processed through an Analog-to-Digital Converter (ADC) to transform the analog signals into digital audio files. Subsequently, the data undergoes exploratory data analysis (EDA) to extract meaningful patterns and features. Finally, these processed data are input into a classification model to identify and classify the bird species.

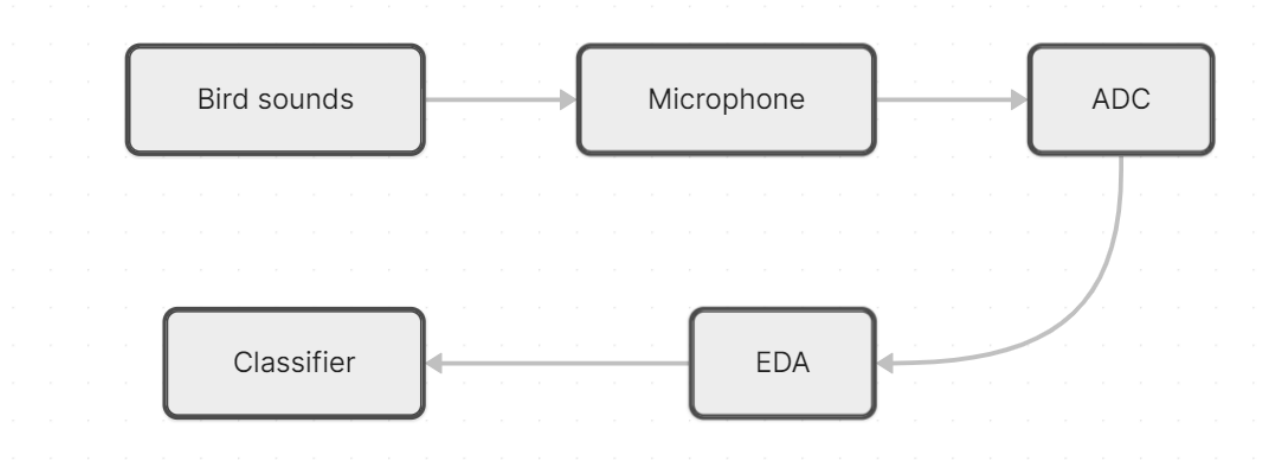


Figure 2.1: Abstraction of the bird species classification

2.3. Data processing

There has been significant research on bird species classification using raw audio signals [2] [3]. However, the authors in [4] argued that mel-spectrogram images yield better performance in preprocessing audio data compared to raw audio signals. The use of mel-spectrogram images continues to improve the accuracy and generalizability of classification models. Mel-spectrograms are particularly useful for bird species classification through audio data due to their ability to effectively represent frequency patterns. By decomposing audio signals into distinct frequency components over time, mel-spectrograms serve as a powerful feature extraction technique [5]. Additionally, mel-spectrograms provide a visually interpretable representation, supporting both model validation and human analysis of acoustic features. Furthermore, their reduced dimensionality

compared to raw waveforms enhances computational efficiency. This method has demonstrated success across various audio classification tasks, making it a reliable choice for bird species identification. Finally, the mel-spectrograms are converted into image format, transforming the task into an image classification problem.

2.4. Classifier

2.4.1. Machine learning

Traditional machine learning (ML) has been a common approach for classification tasks, offering relatively good performance and fast training times when working with small, well-preprocessed datasets containing a limited number of classes *Figure 2.2*. Common classifiers such as Support Vector Machines (SVM), Naive Bayes (NB), and K-Nearest Neighbors (KNN) have proven effective under these conditions. However, as datasets become more complex, contain more noise, high variance, overlapping and involve a larger number of classes *Figure 2.3*, *Figure 2.4*, *Figure 2.5*, the performance of ML classifiers tends to decline significantly. This is because these models require extensive data preprocessing to perform well, making them less suitable for more challenging bird song classification tasks.

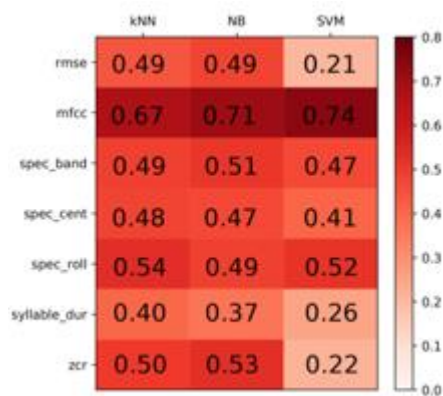


Figure 2.2: F-score for each classifier/feature pair tested and 3 species[6]

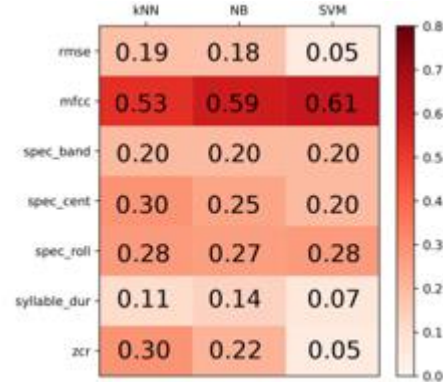


Figure 2.3: F-score for each classifier/feature pair tested and 8 species[6]

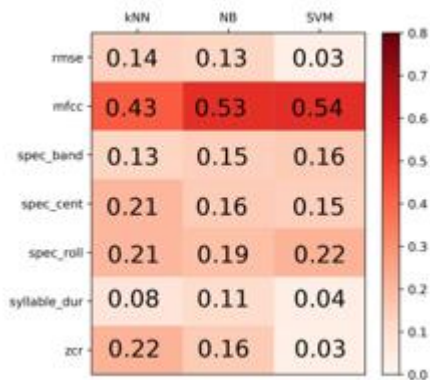


Figure 2.4: F-score for each classifier/feature pair tested and 12 species[6]

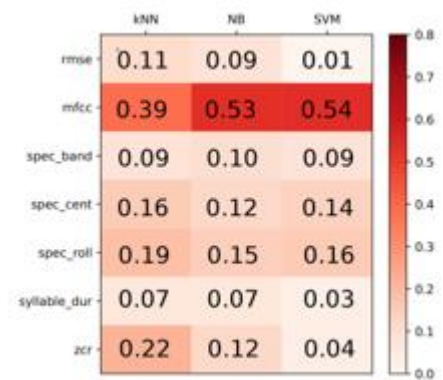


Figure 2.5: F-score for each classifier/feature pair tested and 20 species[6]

2.4.2 Deep Learning

Deep learning techniques, particularly **Convolutional Neural Networks (CNNs)**, have gained significant attention for identifying bird species through their vocalizations. Thanks to their powerful feature extraction capabilities, CNNs can effectively address challenges such as **noise**, **overlapping calls**, and **high variability** in bird songs without requiring extensive filtering or preprocessing steps. Additionally, CNNs perform exceptionally well with large datasets, making them a preferred choice for complex bird song classification tasks.

Recent research has shown how CNNs can improve bird species classification. In [7], researchers used a CNN model with five convolutional layers and one dense layer to classify bird species. They processed the audio data by separating bird sounds from background noise, then created spectrograms and split them into fixed-length parts. This method, tested on the BirdCLEF 2016 dataset, achieved a mean average precision (mAP) score of 0.686, showing the potential of CNNs in this task. Similarly, in [8], a custom CNN model was used on the BirdCLEF 2017 dataset. Researchers used mel-spectrograms created from audio data, added data augmentation, and split the recordings into fixed segments. This model achieved an mAP score of 0.605.

However, while CNNs are good at finding spatial features, they are not designed to handle time-based patterns in sequential data like bird songs. Recurrent Neural Networks (RNNs) are better for this because they can learn patterns that happen over time. Popular types of RNNs like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) work well to understand long-term dependencies and avoid common issues like vanishing gradients. In bird song classification, RNNs are often added to CNNs to help the model learn both the patterns in time and the features from the audio.

Hybrid models combining CNNs and Recurrent Neural Networks (RNNs) have also been explored to better handle the sequential nature of audio data. Studies like [9] [10] focused on CNN + RNN models. In [10], researchers tested advanced image classification models like ResNet18, ResNet50, and VGG16. The highest accuracy of 61.9% was achieved using VGG16. By integrating a hybrid CNN-RNN model, they improved the accuracy to 67%. This study used the Cornell Bird Challenge 2020 dataset, which has 264 bird species, but they only worked with 100 species to handle class imbalance issues.

In [11], researchers used the full Cornell Bird Challenge 2020 dataset and designed a CNN model with residual learning and attention mechanisms to focus on important features. This approach improved accuracy, achieving a maximum of 64.37%.

Here is the summary table compare each method with these challenges: ❌ is bad, ✅ is good and ✅✅ is very good.

	SVM	Resnet	EfficientNet	EfficientNet + GRU	EfficientNet + LSTM
Feature extraction	✗	✓	✓	✓ ✓	✓ ✓
High accuracy (small dataset)	✓	✗	✗	✗	✗
High accuracy (big dataset)	✗	✓	✓ ✓	✓ ✓	✓ ✓
Training time	✓ ✓	✓	✓	✗	✗
Noise handling	✗	✓	✓	✓ ✓	✓ ✓
High variability in Birdsong	✗	✓	✓	✓ ✓	✓ ✓
Short duration-overlapping calls	✗	✓	✓	✓ ✓	✓ ✓

Table 2.1: Summary table compare method with each challenges

III. Proposed method

3.1. Design requirements

The proposed model for classifying bird species based on their songs must meet the following performance requirements:

Classification Performance: The model must correctly classify at least 80% of bird species in the dataset.

Evaluation Metrics: Key metrics, including accuracy, F1-score, precision, and recall, should all exceed 0.8.

Class mean average precision Performance: The class mean average precision (cmAP) across all classes should be greater than 0.7.

Data Handling: The model should effectively handle challenges such as environmental noise, overlapping bird songs, and high variability in vocalizations to ensure reliable predictions in real-world conditions.

Training Time: Training time is not a critical factor; performance is prioritized over speed.

This model is designed to meet the challenges of classifying bird songs in real-world conditions while achieving high performance. The system focuses on being accurate and reliable, even when the input data includes noise, overlapping calls, or differences in bird songs. Since the model is intended for use in research and experimental labs, training time is less important than ensuring high accuracy and robust results. These requirements will be used to evaluate the system during testing to ensure it meets the expectations.

3.2. My proposed method

From the comparison in *Table 2.1*, it is evident that traditional machine learning models, such as SVM, perform well on small datasets but fail to scale effectively with larger and more complex datasets. On the other hand, CNNs and hybrid CNN-RNN models provide better performance but come with significantly higher training times and computational resource requirements. CNNs are excellent for extracting key features from input data, while RNNs are well-suited for modeling time patterns in sequential data. Additionally, CNNs have shown the ability to handle real-world challenges like noise and overlapping calls and have been widely applied in annual bird song challenges, as illustrated in *Figure 3.1*. Incorporating RNNs after CNN layers further boosts performance, although this comes at the cost of increased training time and computational demand.

Given our design requirements, where achieving high performance is the top priority and training time is less critical, we propose using **EfficientNet**, one of the most advanced CNN architectures, combined with **RNNs (LSTM or GRU)**. This choice ensures optimal feature extraction and sequential data modeling to address the challenges of bird song classification in real-world conditions.

Challenge	Rank	Network adopted	Scores	No. of species	Related literature
BirdCLEF2023	1	NFNet/ConvNeXt/ ConvNeXtV2	c-mAP: 0.7639	264	–
	2	EfficientNetV2/ ResNet-34/ EfficientNet-B0/ EfficientNet-B3	c-mAP: 0.7637		–
	3	EfficientNet-B0/ EfficientNetV2	c-mAP: 0.7631		–
BirdCLEF2022	1	EfficientNet-B3/ NFNet	macro F1: 0.8527	113	–
	2	ReNeXt50/ EfficientNet-B0/ EfficientNetV2/ NFNet	macro F1: 0.8438		–
BirdCLEF2021	1	ResNeSt	micro averaged F1: 0.6932	397	–
	2	ResNet-34/ EfficientNetV2	micro averaged F1: 0.6893		–
	3	ResNet-50/ EfficientNet-B2~B7	micro averaged F1: 0.6891		–
	10	ResNeSt-50 EfficientNet DenseNet121	micro averaged F1: 0.6738		Conde et al, 2021
BirdCLEF2020	1	由NAS定义 Built by NAS	c-mAP: 0.128	960	Voelker et al, 2019
	2	Xception	c-mAP: 0.042		Bai et al, 2020
	3	Alexnet	c-mAP: 0.063		Muhling et al, 2020
BirdCLEF2019	1	ResNet/ Inception	c-mAP: 0.356	659	Lasseck, 2019
	2	ResNet/ Inception	c-mAP: 0.160		Koh et al, 2019
	3	Inception-v3	c-mAP: 0.054		Bai et al, 2019

Figure 3.1: BirdCLEF challenge top score in 5 years [12]

EfficientNet is an ideal choice due to its lightweight design and strong performance. One of its key innovations is **compound scaling**, a method that systematically balances three dimensions of a neural network: **width**, **depth**, and **resolution**. Unlike traditional methods that scale only one dimension at a time (e.g., making the network deeper or wider), compound scaling adjusts all three dimensions proportionally to optimize performance while minimizing computational costs.

Width Scaling: Increases the number of channels in each layer to improve the model’s ability to capture more detailed features. This is useful for detecting fine-grained patterns in bird songs.

Depth Scaling: Adds more layers to the model, allowing it to extract increasingly complex features. This helps in identifying subtle variations in bird vocalizations.

Resolution Scaling: Increases the input image size, enabling the model to process higher-resolution mel-spectrograms, which can represent finer details in the audio.

The **compound scaling strategy** combines these adjustments in a balanced way, as shown in Figure 3.2. This approach ensures that the network grows in a coordinated manner, avoiding overfitting or excessive computational requirements. For example, instead of arbitrarily deepening or

widening the network, EfficientNet applies a scaling coefficient to proportionally adjust depth, width, and resolution based on the available resources and the task requirements.

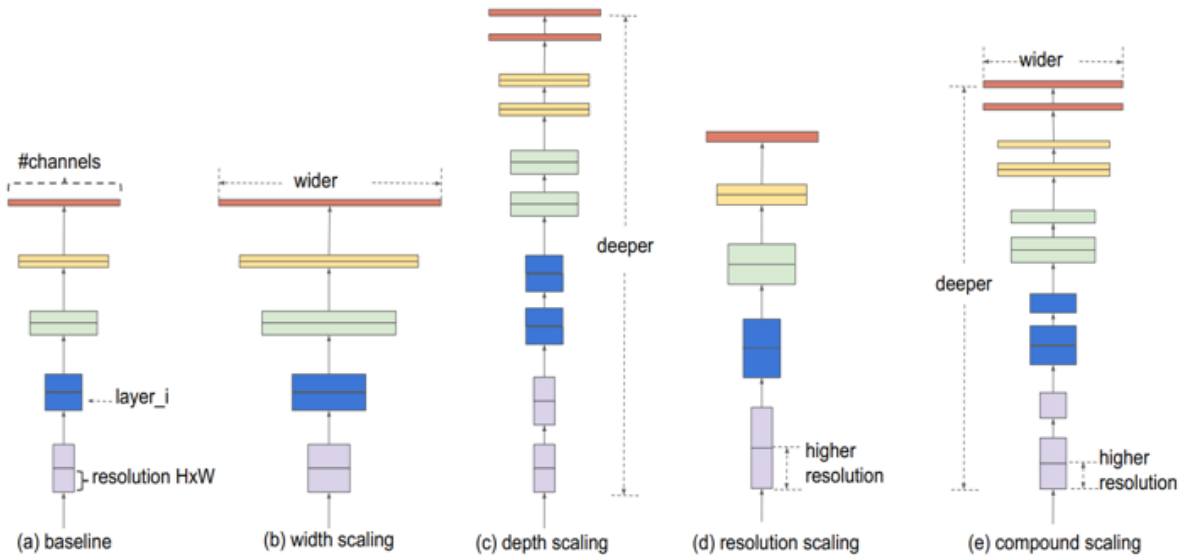


Figure 3.2: Model scaling. a) is a baseline network example; b,d) are conventional scaling that only increase one dimensions of network width, or resolution; e) is compound scaling method that uniformly scales all three dimensions with a fixed ratio[13].

And the benchmark performance of EfficientNet as show in Figure 3.3, which is illustrates the relationship between model size (number of parameters) and ImageNet accuracy for various architectures, highlighting the efficiency of EfficientNet. Notably, EfficientNet achieves superior accuracy with significantly fewer parameters compared to other popular models like ResNet, NASNet, and AmoebaNet. For instance, **EfficientNet-B1** achieves an accuracy of 78.8% with only 7.8 million parameters, outperforming **ResNet-152**, which requires 60 million parameters to reach 77.8% accuracy. As the model scales from EfficientNet-B0 to EfficientNet-B7 using its compound scaling strategy, it consistently achieves higher accuracy while maintaining relatively lower parameter counts compared to competing architectures. For example, **EfficientNet-B7** reaches a remarkable 84.4% accuracy with 66 million parameters, making it both accurate and computationally efficient. This ability to balance model size and performance underscores why EfficientNet is an ideal choice for tasks like bird song classification, where computational resources might be limited, yet high accuracy is essential.

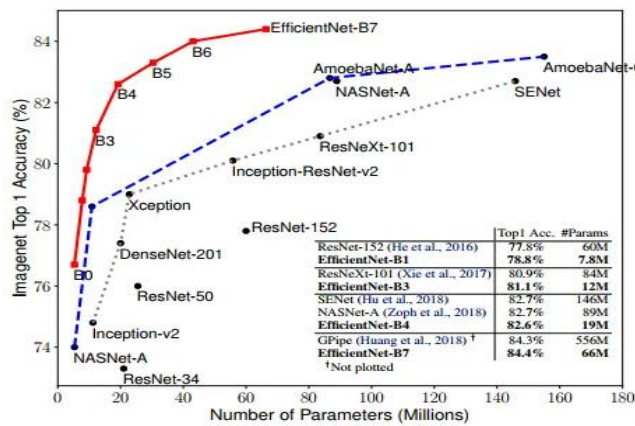


Figure 3.3: Model size vs ImageNet accuracy [13]

The compound scaling method used in EfficientNet is a systematic approach to scaling neural networks by balancing three key dimensions: depth, width, and resolution. Instead of scaling these dimensions independently, the authors introduced a compound coefficient (ϕ) to scale them uniformly under the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, where α, β, γ represent the scaling factors for depth, width, and resolution, respectively. This ensures that the network grows proportionally and remains efficient.

The process starts with fixing $\phi = 1$ and performing a grid search to find optimal values for α, β, γ . For EfficientNet-B0, the best values are $\alpha = 1.2, \beta = 1.1, \gamma = 1.15$. These values are then fixed, and the baseline network is scaled up to create larger models (EfficientNet-B1 to B7) by incrementing ϕ . This approach enables EfficientNet to achieve state-of-the-art accuracy while maintaining a relatively low number of parameters and computational costs compared to traditional CNN architectures, making it highly efficient and effective for tasks like bird species classification.

Recurrent Neural Networks (RNNs) are widely used for processing sequential data like audio, as they are designed to learn temporal dependencies in data. Among RNN variants, **Long Short-Term Memory (LSTM)** and **Gated Recurrent Unit (GRU)** are particularly effective for handling time-series data due to their ability to mitigate the vanishing gradient problem, which often affects standard RNNs. Both architectures achieve this through specialized gating mechanisms that control the flow of information.

LSTM, introduced by Hochreiter and Schmidhuber in 1997, utilizes three gates—input gate, forget gate, and output gate—to regulate the information added, retained, or removed from the memory cell at each time step. The memory cell (C_t) enables the LSTM to capture long-term dependencies, making it highly effective for tasks that require learning patterns over extended time periods, such as bird song classification.

GRU, proposed by Cho et al. in 2014, simplifies the LSTM architecture by combining the forget and input gates into a single update gate and using a reset gate to control how much of the previous hidden state contributes to the current state. This streamlined design reduces the computational complexity while maintaining performance, often making GRUs faster and more resource-efficient than LSTMs.

Both architectures are suitable for bird song classification, where temporal patterns in audio are crucial for identifying different species. By combining EfficientNet with RNNs, the proposed model leverages the feature extraction capabilities of CNNs and the sequential learning strengths of RNNs. This hybrid approach is designed to meet the design requirements, achieving high accuracy while addressing the challenges of real-world bird song classification. Additionally, training time is not a primary concern, allowing the focus to be on optimizing classification performance.

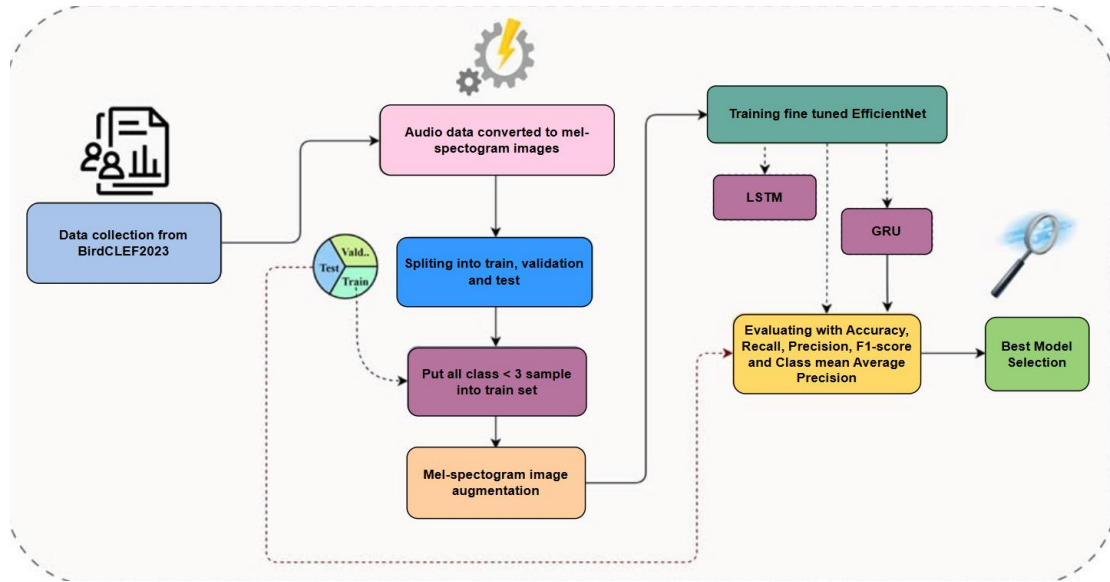


Figure 3.4: Summary of proposed method

The overall structure of the system is illustrated in *Figure 3.4*. The process begins with data from the BirdCLEF 2023 dataset, which is converted into mel-spectrogram images to represent the audio signals visually. Following this, data augmentation techniques are applied to enhance the variability and robustness of the dataset. The dataset is then split into training, validation, and test sets, ensuring that rare classes with only one sample are included in the training set to improve model generalization for these underrepresented categories. The fine-tuning process is conducted using a hybrid model, either EfficientNet + LSTM or EfficientNet + GRU, leveraging their complementary strengths in feature extraction and sequential data modeling. Finally, based on evaluation metrics such as Accuracy, F1-score, Precision, Recall, and Class mean Average Precision (cmAP), the model with the best performance will be selected.

IV. Experimental results

4.1. Data overview and processing

Table 4.1 presents basic statistics about the dataset. The collection includes train audio files containing 16,753 audio signals, each recorded at a sample rate of 32,000 Hz, representing 264 distinct bird species. Along with the audio files, the dataset provides a metadata file, “train_metadata.csv”, which includes detailed information about the training data. A key field in this metadata is the “primary_label”, which identifies the bird species associated with each recording.

Items	Value
Number of bird sounds	16753
Number of bird species	264
Number of sampling rate of audios	32000 Hz

Table 4.1: Basic statistics of dataset

However, the dataset shows an uneven distribution of samples across species, as illustrated in Figure 4.1. Some bird species have many samples, with the maximum being 500, while others are rare, with only one sample available. Additionally, many audio recordings contain background noise, including songs from other bird species. Across the entire dataset, approximately 15,000 noisy bird songs were identified Figure 4.2, which adds complexity to the data preparation process.

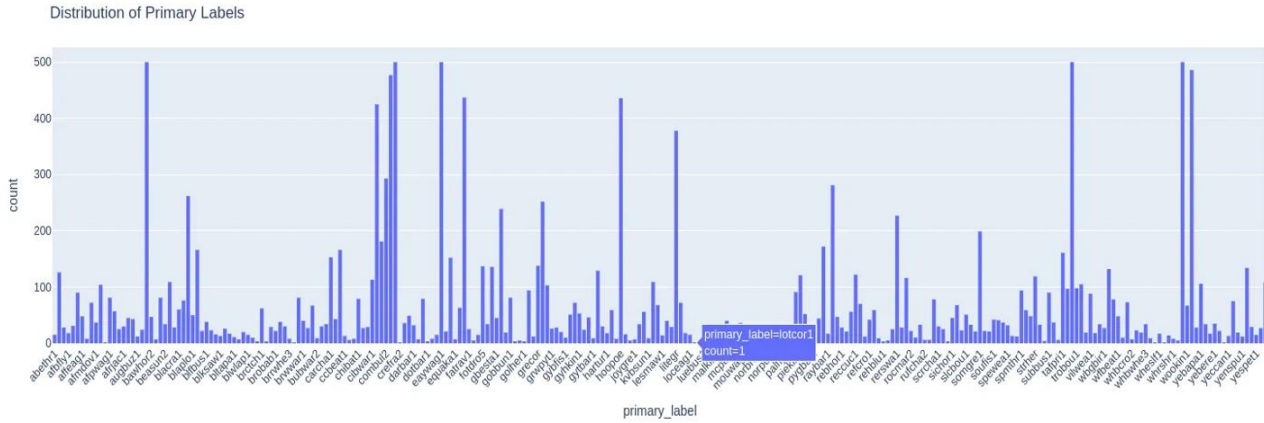


Figure 4.1: Distribution of birds' species data

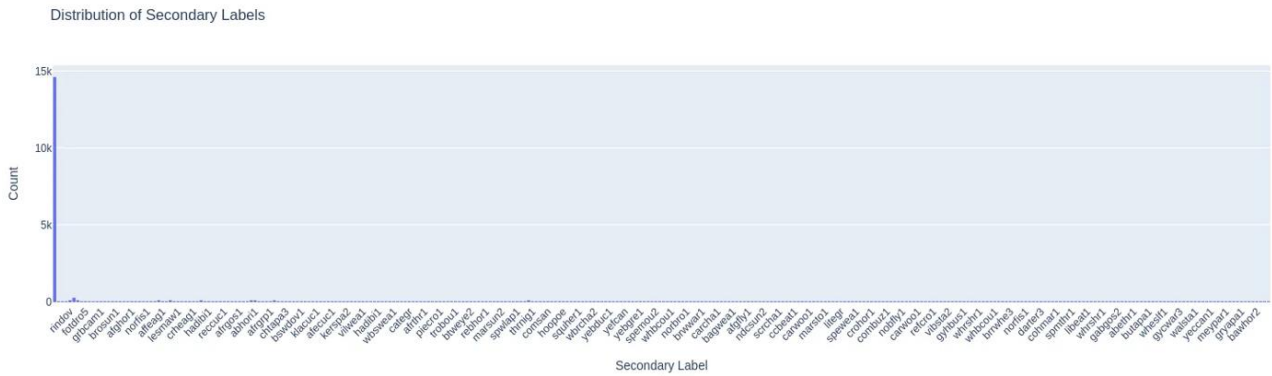


Figure 4.2: Distribution of noise

To prepare the data for training, the following preprocessing steps were carried out:

1. **File Collection:** Audio files and their corresponding labels were collected from the database.
2. **Segmentation:** Each audio file was split into **5-second segments**, and shorter segments (less than 5 seconds) were padded to meet the required duration.
3. **Mel Spectrogram Transformation:** Mel spectrograms were generated for each 5-second audio segment using parameters specified in *Table 4.2*.
4. **Normalization:** The spectrograms were transformed into the decibel scale and normalized to the **uint8 image format** to make them suitable for image-based processing. *Figure 4.3* provides examples of the resulting spectrogram images.
5. **Label Assignment:** A primary label was assigned to each spectrogram image. This label indicates the dominant bird species or sound classification during the given time segment. The connection between the primary label and the corresponding spectrogram is crucial for training and classification, as it forms the basis for supervised learning.

Parameter	Value
Samples between consecutive windows	2048
Number of mel frequency bands	128
Minimum frequency of mel filter bank (Hz)	128
Maximum frequency of mel filter bank (Hz)	500
Centered	False

Table 4.2: Parameters and value for mel spectrogram conversion

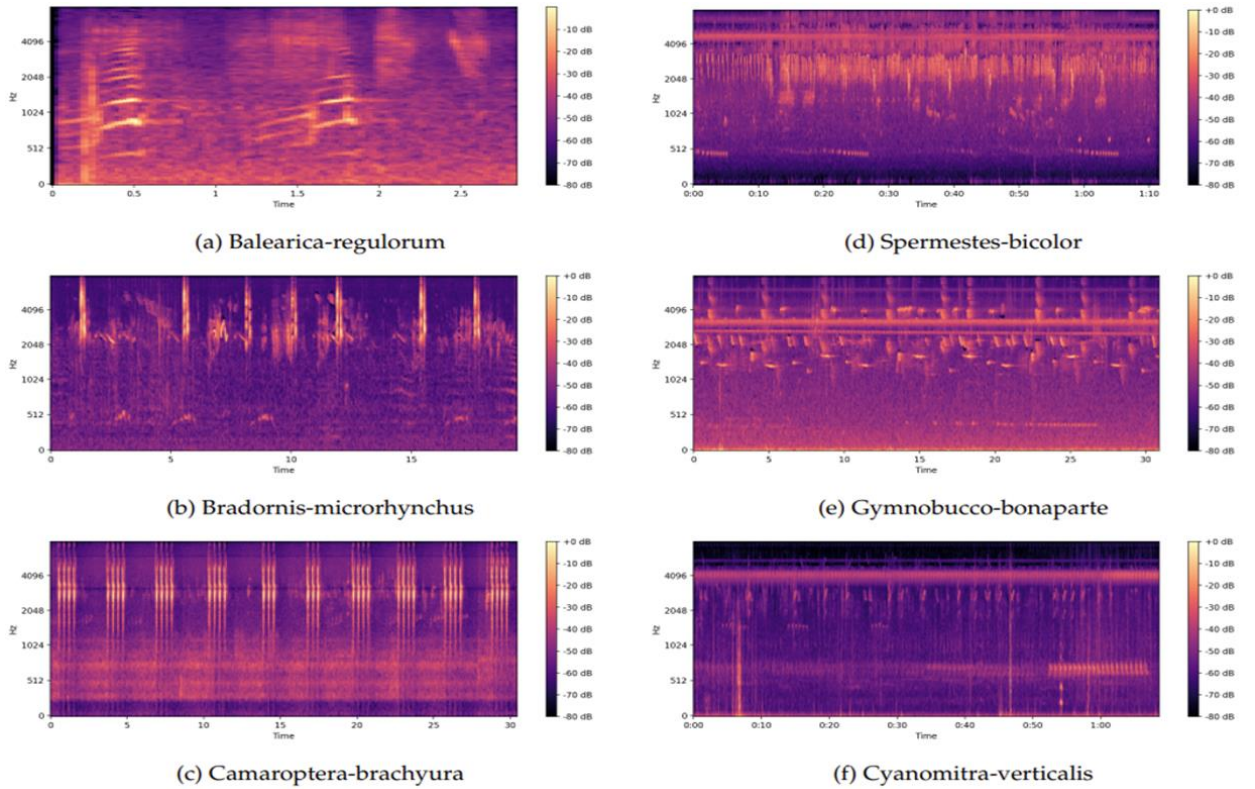


Figure 4.3: Mel-spectrogram images of some bird species audio

After completing the processing steps, the dataset was transformed from audio recordings into approximately 124,000 mel spectrogram images (128, 256), with each image representing a 5-second segment of audio. Primary labels were assigned to each interval, ensuring that each spectrogram was labeled with the predominant bird species within the segment. Padding was applied to shorter segments to meet the minimum duration, and the independent creation of spectrogram images ensured consistency. This transformation effectively converted the bird species classification task into an image classification problem.

To enhance the variability and robustness of the dataset, data augmentation techniques were randomly applied to the spectrogram images with a probability of 0.8. The augmentation techniques used were as follows *Figure 4.4*:

```
RandomBrightness(factor=0.2),
RandomContrast(factor=0.5),
GridMask(ratio_factor=(0.02, 0.05)),
RandomGaussianBlur(kernel_size=3, factor=1),
RandomRowMask(param=5, num_mask=2),
RandomColumnMask(param=5, num_mask=2),
```

Figure 4.4: Augmentation parameter

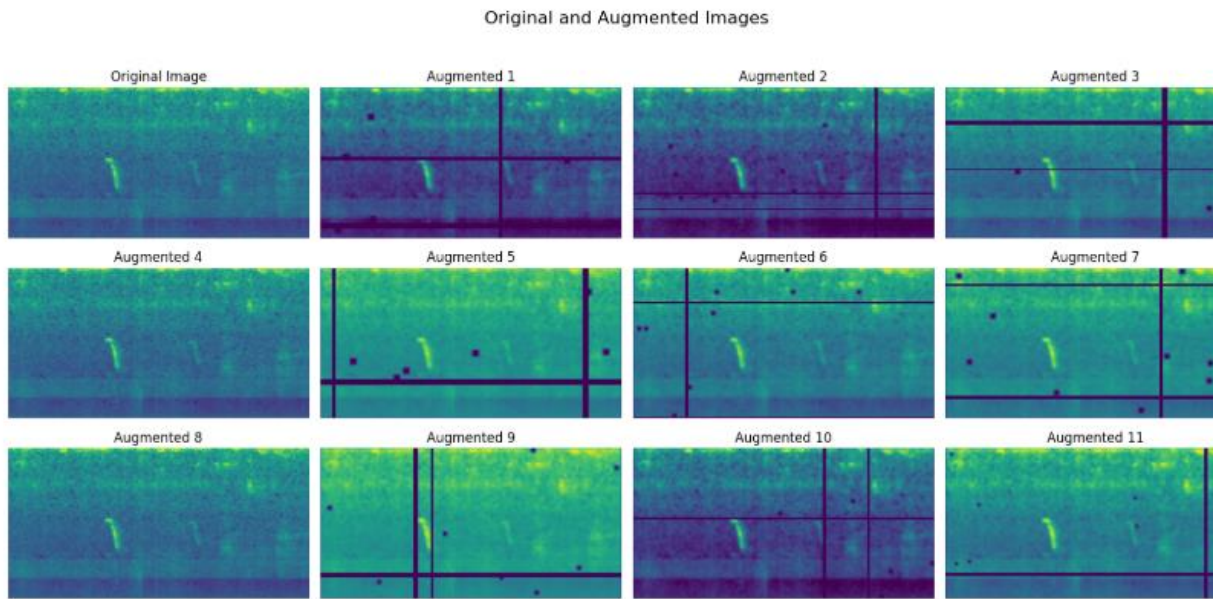


Figure 4.5: Random augmentation in 1 image

As illustrated in *Figure 4.5*, these augmentation techniques generate diverse variations of the original spectrograms, increasing the dataset's size and variability. This process is critical for preventing overfitting and improving the model's ability to generalize across different bird species. The augmented dataset provides a strong foundation for the subsequent training and fine-tuning stages of the classification model.

Next, the dataset was split into training, validation, and test sets with a ratio of 80:10:10, ensuring a balanced distribution of samples. Shuffling was applied (`shuffle = True`) during the splitting process to maintain randomness and prevent bias. For classes with fewer than 3 samples, all instances were

included in the training set to ensure that the validation and test sets had enough samples for meaningful evaluation.

The prepared datasets were then passed through a dataloader with a batch size of 400, facilitating efficient data loading and preprocessing during model training and evaluation.

4.2. Evaluation method

To assess the performance of the bird species classification model, the following metrics were used: **Accuracy**, **Precision**, **Recall**, **F1-score**, and **class mean Average Precision (cmAP)**. Each metric provides valuable insights into different aspects of the model's performance.

Accuracy: measures the percentage of correctly classified samples out of the total samples. It provides a general overview of the model's effectiveness.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN},$$

where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

In this context, Accuracy shows how well the model predicts bird species (and non-bird classes) across the dataset. High Accuracy generally indicates good overall performance. However, Accuracy can be misleading in imbalanced datasets, where some bird species are much more prevalent than others. For example, if the model always predicts the most common species, Accuracy may remain high, but the model fails to classify rare species effectively. Thus, Accuracy alone is insufficient for evaluating models in tasks with significant class imbalance, such as bird song classification.

Precision: focuses on the quality of the model's positive predictions, answering the question: Out of all samples predicted as positive (a specific bird species), how many are truly positive?

$$Precision = \frac{TP}{TP + FP}$$

Precision is crucial for ensuring that the model does not confuse one bird species with another (minimizing false positives). For example, a Precision of 80% for a bird species means that out of all samples predicted as belonging to that species, 80% are correct, while 20% are misclassifications. High Precision indicates that the model is reliable in identifying species without mistakenly predicting incorrect ones. This is particularly important for rare species, where false positives could overwhelm true positives, leading to unreliable classifications.

Recall (Sensitivity): measures the model's ability to identify all true positive samples, answering the question: Out of all the samples that truly belong to a species, how many did the model correctly classify?.

$$Recall = \frac{TP}{TP + FN}$$

High Recall means the model misses fewer true samples (minimizing **false negatives**), making it especially important for ensuring that the model can detect as many bird species as possible.

This is critical in datasets with rare species or noisy samples, where the goal is to capture every possible occurrence of a species. For example, a **Recall of 60%** for a species means the model correctly classified 60% of its actual samples, while 40% were missed. In bird classification, Recall is vital to prevent the model from ignoring rare or underrepresented species.

F1-score: Provides a balance between Precision and Recall by calculating their harmonic mean. It is particularly useful for imbalanced datasets, ensuring a fair evaluation of both metrics.

$$F1 - score = 2 \times \left(\frac{Precision \times Recall}{Precision + Recall} \right)$$

It is particularly useful for evaluating models on imbalanced datasets, where both false positives and false negatives are significant concerns. The F1-score ensures a balanced assessment by penalizing models that perform well on one metric (e.g., high Precision) but poorly on another (e.g., low Recall). For example, a high F1-score indicates that the model achieves both high Precision (few misclassifications) and high Recall (few missed samples), making it an essential metric for bird song classification tasks.

Class mean Average Precision (cmAP): cmAP evaluates the model's performance across all classes by calculating the unweighted mean Average Precision for each class.

$$cmAP = \frac{1}{N} \sum_{i=1}^N AP_i,$$

where AP_i is the average precision for class i and N is the total number of classes.

Unlike traditional metrics like Accuracy, Precision, and Recall, **cmAP focuses on class-level performance**, ensuring that all species, including rare ones with only a few samples, are treated equally. This makes cmAP particularly suitable for datasets like BirdCLEF, where some species are highly imbalanced (e.g., hundreds of samples for common species but only 1-2 samples for rare ones).

The difference between **mAP (mean Average Precision)** and **cmAP** lies in how they handle class imbalance:

mAP (mean Average Precision): Averages the AP across all samples, giving more weight to classes with more samples. As a result, common species dominate the evaluation, and rare species may not be adequately represented.

cmAP (class mean Average Precision): Averages the AP across all classes equally, regardless of the number of samples. This ensures that rare species are not overshadowed by common ones, providing a fairer evaluation.

For example, if a rare species has only one sample and the model misclassifies it, mAP might not drop significantly because the rare species contributes minimally to the overall score. In contrast, cmAP would highlight this misclassification, reflecting the model's inability to generalize across all species.

And padded cmAP is a modified version of the cmAP metric designed to address challenges with imbalanced datasets, particularly for categories with few or no true positives. It works by adding five synthetic true positives (padding) to both the ground truth and predictions for each category. This

ensures that even rare or absent categories can be evaluated, stabilizing the metric and reducing fluctuations caused by limited samples. Padding does not significantly affect common categories but provides a fair and consistent evaluation framework for all categories, making the metric more robust and equitable.

By using **padded cmAP** with padding = 5, the evaluation becomes both fair and robust, ensuring that rare species are not ignored while preventing excessive penalization of the model.

4.3 Implementation

After converting the audio data into mel spectrogram images and defining the metrics for evaluation, the training process for the bird species classification by song project was initiated. The training task was formulated as a supervised multiclass classification problem, and the CrossEntropyLoss function was employed as the objective metric. This loss function is suitable for such tasks as it measures the performance of a classification model whose output is a probability distribution over multiple classes.

The training was conducted using the Adam optimizer, starting with an initial learning rate of 1×10^{-4} . Adam was selected for its adaptive learning rate capability, which facilitates efficient and stable convergence by dynamically adjusting the learning rate based on the gradient changes during training. This choice ensures that the model parameters are effectively optimized throughout the training process.

Training Environment and Settings

The experiments were executed in a Kaggle environment, leveraging two NVIDIA T4 GPUs. The PyTorch framework was utilized for model implementation, with the DataParallel wrapper enabling parallel training across the two GPUs. This setup ensures that the computational workload is distributed efficiently, reducing training time while maintaining performance.

A batch size of 400 was selected after careful consideration. While smaller batch sizes can lead to slower convergence and potentially prolonged training times, they are sometimes necessary for managing memory constraints in larger models. On the other hand, excessively large batch sizes can cause instability in GPU memory usage, risking training failures. By selecting this batch size, a balance was achieved between computational efficiency and model stability, ensuring smooth and effective training.

In addition, dropout was incorporated to mitigate overfitting, and an early stopping mechanism was employed with a patience value of 20 epochs. The early stopping criterion monitors the validation loss and halts the training process when no significant improvement is observed over the defined patience period. This approach not only conserves computational resources but also helps in selecting the best-performing model checkpoint.

Comparing Hybrid Architectures:

The study aimed to compare two hybrid architectures: EfficientNet + LSTM and EfficientNet + GRU. Data augmentation was applied to the mel spectrogram images as it significantly improved results compared to training on unaugmented data. Consequently, only augmented data results are reported in this comparison.

EfficientNet-B0 embedded with LSTM

EfficientNet-B0 was chosen as the base model due to limited computational resources. Using larger models with the dataset of 124k images (128x256) would have required reducing the batch size and significantly increasing training time, which was not feasible given Kaggle’s training session limit of approximately 43,000 seconds per run. EfficientNet-B0 provided a balanced solution, delivering satisfactory results within these constraints. The model was modified by removing the final classification head, retaining only the feature extraction layers to focus on spatial features of the mel spectrograms. These extracted features (output size of 1280) were then processed by an LSTM layer to capture temporal information, with a sequence length of 1 as each spectrogram frame was treated as a single sequence. Dropout was applied to prevent overfitting, followed by a linear layer with a hidden size of 256. The training process reached 80 epochs, during which early stopping was also triggered, indicating no significant improvement in validation loss. The model’s performance at epoch 80 is detailed in *Table 4.3*, and the accuracy and loss curves in *Figure 4.6* show that the model learned well on the training set but performed less effectively on the validation set. The validation loss stopped decreasing, which suggests the model was close to overfitting and unable to improve further on unseen data.

<i>Train</i>		<i>Valid</i>		<i>Test</i>				
<i>Accuracy</i>	<i>Loss</i>	<i>Accuracy</i>	<i>Loss</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	<i>cmAP</i>
98.225%	0.074	86.02%	0.751	87.06%	0.835	0.813	0.818	0.778

Table 4.3: performance of efficientb0+lstm

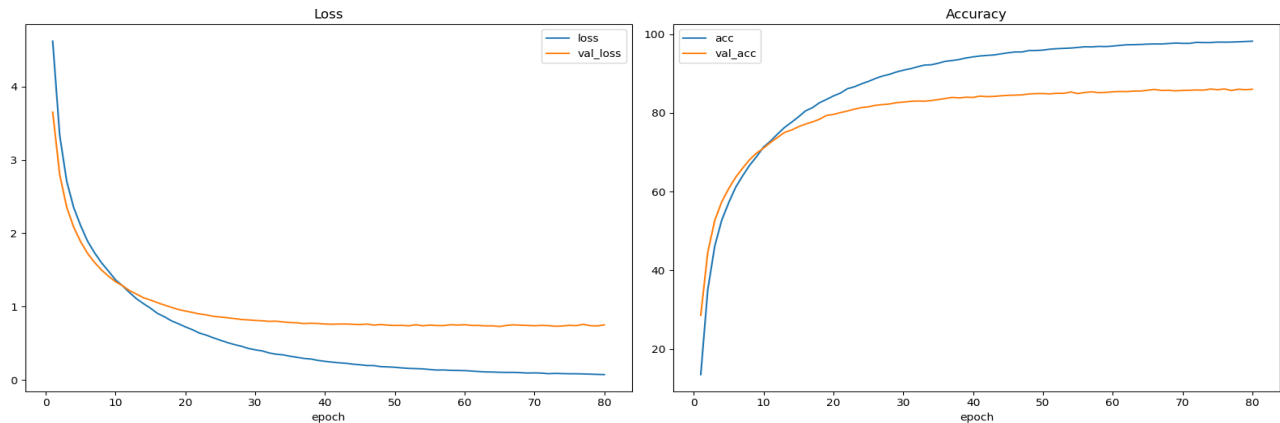


Figure 4.6: Loss and accuracy curve of train and validation set on EfficientNet-B0 + LSTM

EfficientNet-B0 embedded with GRU

In our second model, EfficientNet-B7 with GRU was implemented. The model was trained for 80 epochs, resulting in *Table 4.4*. In *Figure 4.7*, epoch-wise train/valid loss and accuracy scores are depicted. The left side of the figure shows the train/valid loss decreasing over the epochs. On the contrary, the right side shows the train/valid accuracy increasing over the epochs. After completing 80 epochs, the model loss approached convergence, but a slight indication of overfitting became apparent. Therefore, it was decided to conclude the training at that stage.

Train		Valid		Test				
<i>Accuracy</i>	<i>Loss</i>	<i>Accuracy</i>	<i>Loss</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	<i>cmAP</i>
98.31%	0.07	86.16%	0.74	87.16%	0.829	0.795	0.80	0.770

Table 4.4: performance of efficientb0+gru

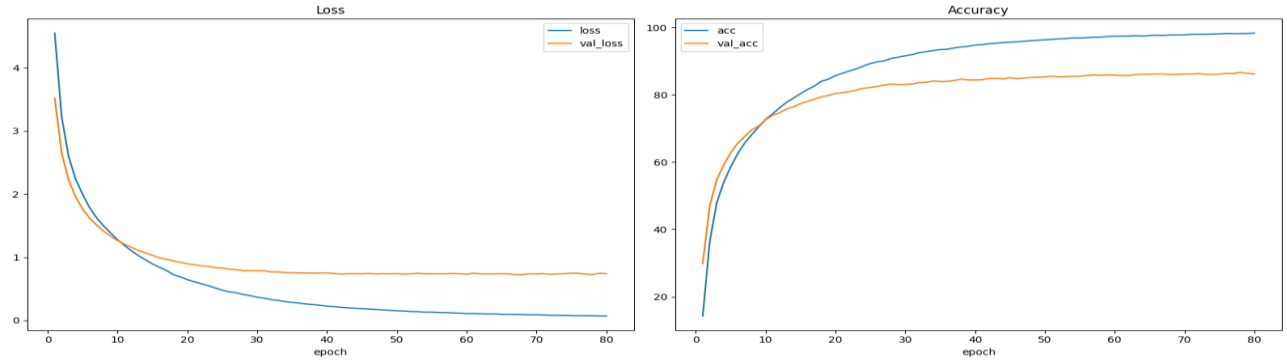


Figure 4.7: Loss and accuracy curve of train and validation set on EfficientNet-B0+GRU

Technique	Accuracy	Precision	Recall	F1-score	Class mean average precision
EfficientNet-B0 + LSTM	87.06%	0.835	0.813	0.818	0.778
EfficientNet-B0 + GRU	87.16%	0.829	0.795	0.80	0.770

Table 4.5: Summary performance of each model

After comparing the performance of LSTM and GRU models, we observed no significant difference between the two. However, LSTM showed slightly better results, making it the preferred choice for further visualization. Thus, we selected the EfficientNet-B0 + LSTM architecture for the next steps.

After testing the model on the test set using various metrics, we visualized the errors to understand where they usually occur. We identified the top 10 species with the highest number of errors, as shown in *Figure 4.8*. Next, we checked whether these errors came from common or rare classes. Surprisingly, the top 10 species with the most errors had a very large number of samples, ranging from 250 to nearly 500 samples *Figure 4.9*. This suggests that the high diversity within these species makes it difficult for the model to distinguish them compared to classes with fewer samples.

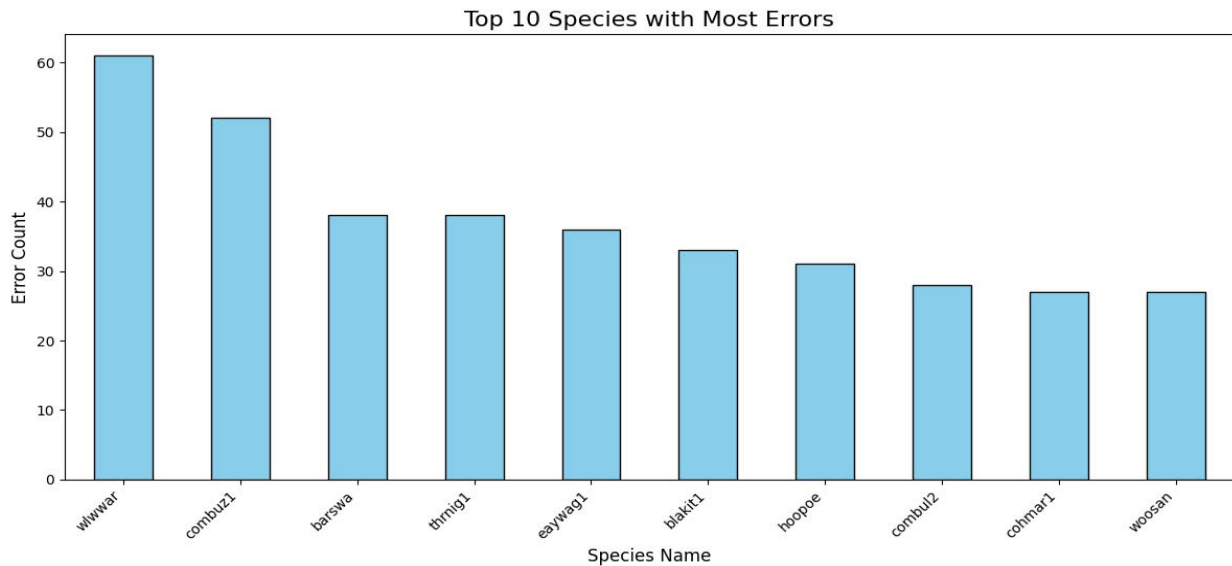


Figure 4.8: Top 10 species have most error

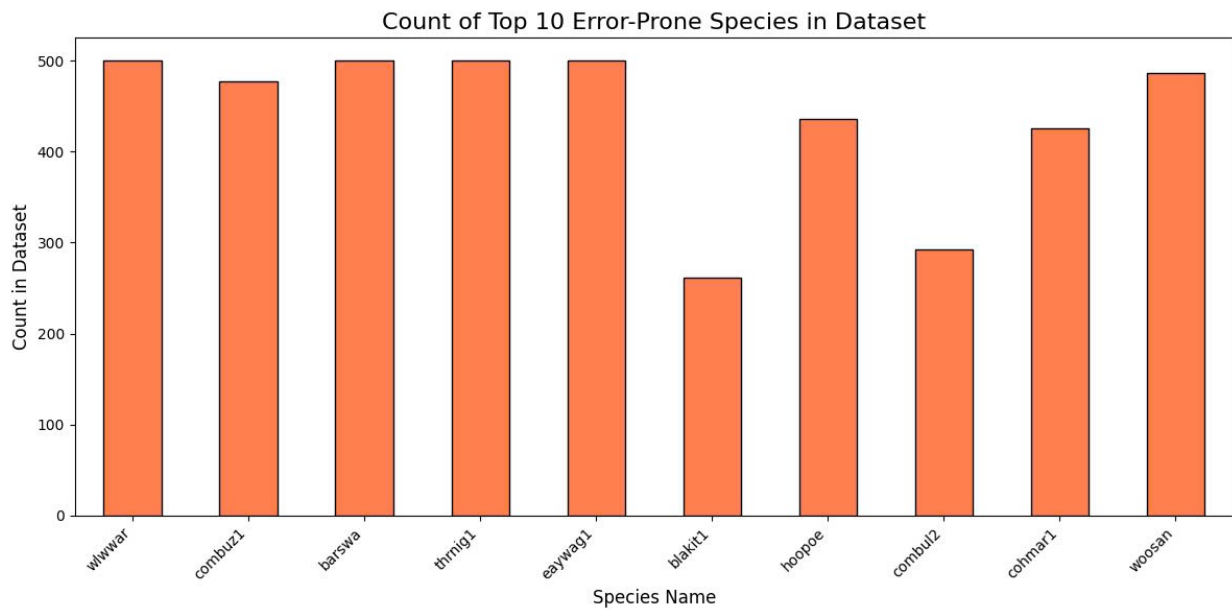


Figure 4.9: Count of Top 10 species have most error

Limitation

The training results and performance suggest that the model is likely experiencing overfitting. This is partly due to the high variability within the dataset, which makes it difficult for the model to learn effectively, as most errors in the test set are concentrated in common classes *Figure 4.9*. Additionally, the dataset's imbalance poses a significant limitation, as some rare classes are included only in the training set, making it impossible to evaluate the model's ability to generalize to these classes. The use of the EfficientNet + LSTM architecture, while effective, is computationally intensive and limited by training time and hardware constraints. This restricts the ability to experiment with more complex techniques or larger models. Furthermore, the current dataset is not sufficiently large or diverse for the bird species classification task, highlighting the need for a more extensive and balanced dataset to achieve better model performance.

V. Conclusion

This study presents an audio-based bird species classification system specifically designed for Eastern African birds, achieving an accuracy of 87.06% using the BirdCLEF 2023 dataset. The system includes two key components: a mel spectrogram generator and a hybrid deep learning classifier. The mel spectrogram generator converts bird vocalization recordings into mel spectrogram images, which are then analyzed by the classifier. The classifier combines the strengths of CNNs to capture local patterns and RNNs to understand temporal dependencies in the audio data. By applying augmentation techniques and testing various deep learning architectures, the system improves accuracy and contributes to bird conservation, environmental monitoring, and wildlife research.

The system meets the design requirements with satisfactory performance. However, it still struggles to clearly classify both common and rare bird species.

Future work will focus on enhancing classification accuracy through more effective augmentation techniques. Exploring alternative deep learning architectures could further improve the system's performance. Additionally, testing the model on other bird species and under different environmental conditions will help ensure its versatility and generalizability. Expanding the dataset to include more samples and reduce class imbalance will also be a priority to support better model training and performance.

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